

Lab: Planning, Imagination, and Future Thinking

Learning Goal: Analyze how agents plan ahead through simulation, habits, and hierarchical goal decomposition.

Part 1: Planning as Inference

Scenario: You're planning a surprise birthday party for a friend next Saturday.

1. List the subgoals you need to achieve (venue, food, guests, decorations, timing).
2. For each subgoal, describe the uncertainty you need to resolve (epistemic actions).
3. For each subgoal, describe the actions you need to take to achieve preferred outcomes (pragmatic actions).
4. How does your plan change if you learn your friend is planning to leave town Friday night?

Write your response here

Part 2: Timescale Analysis

Learning Goal: Explore planning across multiple timescales.

Exercise: Pick a goal that requires planning at multiple timescales (e.g., graduating from college, running a marathon, learning a language).

Timescale	Goal	Actions	How do you evaluate progress?
Years			
Months			
Weeks			
Days			
Minutes			

Fill in the table. How do the higher-level goals constrain the lower-level ones?

Write your response here

Part 3: Habits vs. Deliberate Plans

Learning Goal: Distinguish habitual from deliberate action selection.

Exercise: For one typical day, list:

1. **Five habitual actions** (things you do without deliberate planning)
2. **Five deliberate actions** (things that required active planning/decision-making)

For each habitual action, answer: What would cause you to switch from habit to deliberate planning? (e.g., What if your usual route to class was blocked?)

Write your response here

Part 4: Counterfactual Reasoning

Learning Goal: Practice imagining alternative scenarios and their consequences.

Exercise: Think of a significant decision you made in the past year.

1. What did you decide?
2. Generate two counterfactual alternatives: "What if I had chosen X instead?" "What if I had chosen Y?"
3. Simulate the likely consequences of each alternative using your generative model.
4. How confident are you in your simulations? What sources of uncertainty make counterfactual reasoning difficult?

Write your response here

Part 5: Course Integration Reflection

In 200 words, trace the full Active Inference story from Module 01 to Module 08. How does a system become an agent, perceive, think, act, learn, communicate, and plan — all through the same underlying principle of minimizing free energy / prediction error?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Goal decomposition	Planning as inference
2	Multi-timescale thinking	Deep temporal models
3	Self-observation	Habits vs. deliberation
4	Counterfactual reasoning	Imagination as offline inference
5	Integrative reflection	Full Active Inference framework